

Welcome to CS4610 - Programming Languages

Please send course-related emails to the course email address:

cs4610@cshelpdesk.atlassian.net (<mailto:cs4610@cshelpdesk.atlassian.net>)

In any emails that you send, please include your name and computing ID.

Weekly TA Office Hours:

TA Office Hours will be hosted by graduate students and will start on Thursday (2/1). These office hours will be held in Rice 442. There will be one CS4610 TA present during these scheduled hours. There will likely be other courses holding office hours in this room at the same time.

Monday: 11:00-2:00 and 2:30-5:30

Tuesday: 10:00-12:00

Wednesday: 11:00-2:00

Thursday: 10:00-12:00

Friday: 11:00-1:00

Prof. Pettit's office hours are held in his office in Rice 212. As these office hours are shared with my other course (CS3100, 250 students), sometimes the number of students attending can be quite large. I generally will invited multiple students into my office at a time. If you need to speak privately with me, please let me know. If there is another student or students in my office and you are waiting outside, please let me know that you are waiting.

Monday: 2:30-4:00

Friday: 2:30-4:00

Books:

No course textbook is required. A few good reference books are: Modern Programming Languages: A Practical Introduction 2nd Edition by Adam Brooks Webber; and Concepts of Programming Languages 11th Edition by Robert W. Sebesta.

Course Description

Presents the fundamental concepts of programming language design and implementation. Emphasizes language paradigms and implementation issues. Develops working programs in languages representing different language paradigms. Many programs oriented toward language implementation issues. *(from SIS)*

As computer scientists, we use programming languages on an almost daily basis but rarely consider how the text we write is translated and executed by a computer. Additionally, our choice of language significantly influences the way in which we represent a problem. Different programming languages are often built for different programming paradigms. By adopting a given language, you are constraining the way in which you can think about a particular problem and how to solve it.

We will study how programming languages are designed and implemented. We will discuss such topics as variable bindings, control flow, types, subtyping, operational semantics, and method invocations. While the development of practical skills is not a major goal of this course, I expect that your practical programming skills improve indirectly. We will also discuss the underlying theory of these topics.

In previous computer science courses, you developed proficiency programming using imperative and object-oriented paradigms (most likely in [Java](https://en.wikipedia.org/wiki/Java_(programming_language)) , [Java \(programming language\)](https://en.wikipedia.org/wiki/Java_(programming_language))), C/C++, and Python). This course will expose you to two different programming paradigms and give you experience with a language associated with each. For developing understanding of the functional programming paradigm, we will be using OCaml. For developing understanding of the declarative/logical programming paradigm, we will be using Prolog.

Goals for Student Competency

At the end of this course, you will be able to:

1. Demonstrate knowledge of fundamental terms and concepts associated with most programming languages.
2. Paraphrase core issues and concepts in programming language design and implementation.
3. Identify and contrast the 3 major programming language families.
4. Demonstrate increased breadth and flexibility with regard to programming languages.
5. Read and write programs of easy to moderate difficulty in a functional language (OCaml).
6. Read and write programs of easy to moderate difficulty in a logical language (Prolog).

Requirements to take this course

1. Course prerequisite: Either (CS2100 and CS2120 with grade of C- or higher) or (CS 2150 with grade of C- or higher).
2. You have mastered the material in the courses leading up to, and including the prerequisite courses.
3. Knowledge of object-oriented and imperative programming in Java.

Evaluating Your Progress in the Course

Weekly Homework Assignments (10%): These will be assigned starting the second week of class. Some assignments will be questions styled after textbook chapter review questions. Other assignments will primarily be programming. Assignments will be submitted to the course website unless otherwise noted. If you are struggling to answer certain questions, make a reasonable attempt as partial credit will usually be given.

Portfolio 1 (15%): The instructor will meet with each team in-person to review all assignments from the first half of the semester. Group members will be asked questions to show mastery of the material in the assignments.

Portfolio 2 (15%): The instructor will meet with each team in-person to review all assignments from the second half of the semester. Group members will be asked questions to show mastery of the material in the assignments.

Project1 - Language Research (15%): Your team will be assigned a programming language of interest to this course. You will do research on the why and how of the language, as well as writing some code. You will write up a brief paper on the language as well as create a short presentation for classtime. More details about this assignment will be given when released.

Midterm Exam (20%), Final Exam (25%): These will be paper-and-pencil style exams and will be taken individually. The midterm will cover content from the first half of the semester. The final exam will be comprehensive. You are not allowed to use your textbook, your own notes, or any other materials for the exams. The dates for all exams will be announced in class and on the course website. These exams are designed to assess the content knowledge, skills and concepts you learn during the semester. Each questions will be a variety of short-answer, applied concept, and multiple-choice questions. Prior to each exam, we will look at sample questions to give you familiarity with the structure of the exam.

Communications

To facilitate consistency in communications of course requirements, assignments, and general

information, your instructor will regularly post announcements, assignments, and grades online through the use of the Canvas course website. There will be links from the course website to other relevant resources such as Gradescope and Piazza (https://piazza.com/virginia/spring2024/cs_4610001  https://piazza.com/virginia/spring2024/cs_4610001). Your UVA email address will also be used for communication and you are responsible for checking your UVA email address periodically. Any significant changes to this syllabus will be posted as announcements to the course website. There may be times when I am not online during “office hours” due to various circumstances (sick children, “surprise” meetings, travel, etc.). I will try to post any changes ahead of time.

Asking the instructor questions – Please don’t wait to ask questions until the final day before a major assignment is due. Given the instructor’s work schedule, please allow one work day (generally 24 hours) for a response to questions. If you are asking for help on an assignment that is due in less than 24 hours, you likely won’t receive a response before the due deadline. Please send course-related emails to the course email address (cs4610@cshelpdesk.atlassian.net) (<mailto:cs4610@cshelpdesk.atlassian.net>). In any emails that you send, please include your name and computing ID.

Class Management & Policies

Homework/Project Assignments

- All assignments will be announced in class and posted on Canvas. All assignments will be submitted through the course website unless otherwise noted.
- Homework assignments are done with your group. One member from the group submits the assignment.

Professionalism

Students and staff are all expected to treat each other with respect. This includes, but certainly is not limited to:

- Inappropriate behavior during class (keep humor respectful and PG-rated).
- Disrespectful language
- Promptness for all deadlines and class meetings
- Quality work

Students can and will be penalized for unprofessional behavior.

Regrade Requests

After assignments have been graded and posted, it may be that you would like to discuss the points you earned on a certain part of the assignment. This is considered a regrade request. Unless indicated otherwise, all regrade requests must be made within **1 week** of the assignment being returned to the student. Only submit a regrade request if you feel there is a grading error.

Regrade requests have non-trivial cost to the course staff. To reduce trivial regrade requests, there may be a professionalism penalty given to the assignment if the regrade request is frivolous. Also realize that when regrading an assignment, any other grading errors that are noticed will also be corrected, for better or worse.

Cheating/Plagiarism Penalty

If course staff detect cheating, plagiarism, or other dishonest or unprofessional behavior, they may impose any penalty up to and including a failing grade in the course. This is independent of and in addition to the operations of the Honor Code. The exact penalty will be determined based on the offense. Cite any sources that you use.

Recording of classroom activities

We will be recording every live lecture in order to accommodate students who cannot attend. Because lecture includes fellow students, you and they may be personally identifiable on the recordings. These recordings may only be used for the purpose of individual or group study with other students enrolled in this class during this semester. You may not distribute them in whole or in part through any other platform or to any persons outside of this class, nor may you make your own recordings of this class unless written permission has been obtained from the Instructor and all participants in the class have been informed that recording will occur. If you want additional details on this, please see Provost Policy 008. If you notice that we have failed to activate the recording feature, please remind us!

Students with disabilities or learning needs

It is our goal to create a learning experience that is as accessible as possible. If you anticipate any

issues related to the format, materials, or requirements of this course, please meet with me outside of class so we can explore potential options. Students with disabilities may also wish to work with the Student Disability Access Center to discuss a range of options to removing barriers in this course, including official accommodations. Please visit their website for information on this process and to apply for services online: sdac.studenthealth.virginia.edu. If you have already been approved for accommodations through SDAC, please send me your accommodation letter and meet with me so we can develop an implementation plan together.

Discrimination and power-based violence

The University of Virginia is dedicated to providing a safe and equitable learning environment for all students. To that end, it is vital that you know two values that we and the University hold as critically important:

- Power-based personal violence will not be tolerated.
- Everyone has a responsibility to do their part to maintain a safe community on Grounds.

If you or someone you know has been affected by power-based personal violence, more information can be found on the UVA Sexual Violence website that describes reporting options and resources available - www.virginia.edu/sexualviolence (<http://www.virginia.edu/sexualviolence>).

As your instructors, and as people, know that we care about you and your well-being and stand ready to provide support and resources as we can. As faculty members, we are considered responsible employees, which means that we are required by University policy and federal law to report what you tell us to the University's Title IX Coordinator. The Title IX Coordinator's job is to ensure that the reporting student receives the resources and support that they need, while also reviewing the information presented to determine whether further action is necessary to ensure survivor safety and the safety of the University community. If you wish to report something that you have seen, you can do so at the Just Report It portal (<http://justreportit.virginia.edu/> (<http://justreportit.virginia.edu/>)). The worst possible situation would be for you or your friend to remain silent when there are so many here willing and able to help.

Religious accommodations

It is the University's long-standing policy and practice to reasonably accommodate students so that they do not experience an adverse academic consequence when sincerely held religious

beliefs or observances conflict with academic requirements.

Students who wish to request academic accommodation for a religious observance should submit their request in writing directly to me as far in advance as possible. Students who have questions or concerns about academic accommodations for religious observance or religious beliefs may contact the University's Office for Equal Opportunity and Civil Rights (EOCR) at UVAEOCR@virginia.edu (<mailto:UVAEOCR@virginia.edu>) or 434-924-3200.

Research

Your class work might be used for research purposes. For example, we may use anonymized student assignments to design algorithms or build tools to help programmers; use data collected from course support tools to better understand and seek to improve student engagement and learning; etc. Any student who wishes to opt out can contact the instructor or TA to do so after final grades have been issued. This has no impact on your grade in any manner.

Notes on Learning

Learning new material can be difficult and time consuming. The general expectation at UVA is that you will spend two hours outside of the classroom for every hour spent inside the classroom for a given course. Some courses may expect more, some less. If you find yourself struggling in this course and you are not spending six hours a week outside of the classroom attempting to understand the material, working on homework problems, etc., then spending more time is the first issue that we will discuss. Spending adequate time, however, may not directly relate to your final course grade. Your final course grade is a reflection of how well you demonstrated that you mastered the material. Because of this, it is important that you are spending your study time outside of class effectively.

The ways in which you learned material in the past may not necessarily work in this course or other future courses. Your instructor has done research in how best to help students learn. One of the better books that I have read include "Make It Stick: The Science of Successful Learning" by Peter Brown and Henry Roediger. Another good book that I refer to is "Why Don't Students Like School" by Daniel Willingham. During this course, I will occasionally take time out from course content to speak specifically about learning and what has been shown to be successful with evidence-based research.

Empirical evidence-based research has demonstrated that traditional note-taking (using a writing

instrument such as a pencil or pen) increases learning significantly better than note-taking by means of typing. For this reason, you are asked to take notes with a pencil or pen, instead of typing. If there is a compelling reason that you cannot do this, please come speak with me.

This Syllabus

This syllabus is to be considered a reference document that can and will be adjusted through the course of the semester to address changing needs. This syllabus can be changed at any time. Any major changes will be posted as announcements on the course website. It is up to the student to monitor announcements for any changes. Final authority on any decision in this course rests with the instructor, not with this document.